Fiscal Year:	FY 2021	**Task Last Updated:**	FY 02/03/2022
PI Name:	Wang, Clay Ph.D.		
Project Title:	Characterization of ISS Microorganisms That Assist in the Decomposition of Complex Organic Matter During Spaceflight		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	90089-5305	**Congressional District:**	37
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2018 Space Biology (ROSBio) NNH18ZTT001N-FG. App B: Flight and Ground Space Biology Research
Start Date:	09/01/2019	**End Date:**	08/31/2022
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA KSC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	February 2022 report: No change in personnel		
COI Name (Institution):	Singh, Nitin Ph.D. (NASA Jet Propulsion Laboratory) Venkateswaran, Kasthuri Ph.D. (NASA Jet Propulsion Laboratory)		
Grant/Contract No.:	80NSSC19K1501		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Title: Characterization of ISS microorganisms that assist in the decomposition of complex organic matter during spaceflight. Key objective of project: A broad array of experiments are performed continuously on International Space Station (ISS). These experiments generate waste matter that needs to be disposed of safely and cost-effectively, e.g., dead plant matter generated from the experiments carried out to check for the effect of microgravity on the development of roots and cells on plant seedlings. Filamentous fungi are used industrially as sources of enzymes to break down complex sugars from plant sources. The Wang group at the University of Southern California and Venkateswaran Group at Jet Propulsion Laboratory (JPL) have sent filamentous fungi on two different spaceflight projects (NASA Micro-10) and (CASIS-BRIC-NP). Our analysis of these flight samples compared to ground controls have shown differential expression particularly of genes related to carbohydrate metabolic processes. In addition, Microbial Tracking (MT-1) project isolated 202 microbial strains (138 bacteria and 64 fungi) and archived at JPL will be utilized to identify the microbial culture or consortium to degrade plant organic matter. This finding provides further justification that filamentous fungi and some unique bacterial strains can potentially assist in the decomposition of complex organic matter during spaceflight. The proposed investigation is a ground-based project to characterize microorganisms isolated from the ISS in the degradation of plant compost materials. We will utilize various simulated gravity systems available at JPL and NASA to understand the mechanism of how ISS microorganisms assist in the decomposition of complex organic matter during spaceflight. The objectives of the project and methods used are: 1) Identify the ISS microbial species isolated during MT-1 project which can degrade plant complex organic matter/ utilize complex sugar sources. 2) Genetically engineer up to five MT-1 microbial species (single or symbiotic combination of bacteria/fungi) down selected from objective 1 using CRISPR/Cas9 technology for creating mutants that are efficient in degrading complex organic matter. 3) Examine the proteomic profile of filamentous fungi and bacterial mutants created in objective 2 and wild type strains using simulated microgravity devices in comparison to ground control. 4) Examine the differential expression of carbohydrate degradation genes and their regulatory genes using simulated microgravity devices in comparison to ground control of mutants and wild type strains using whole genome sequencing and whole genome mRNA sequencing. Significance of proposed work: The proposed project addresses subtopic Pl-C ("Study changes in microbial populations from plants and plant growth media") experiments that demonstrate that symbiotic microbes can assist in the decomposition of complex organic matter during spaceflight. The project will not only provide insights into the mechanisms of microbial degradation of complex organic matter during spaceflight but aid in developing practical measures for mitigating risk to human health and ISS environment through waste disposal. The data generated will augment the NASA GeneLab database, which provides NASA with the statistical confidence to characterize microbes associated with closed habitation and crew health maintenance. The analysis proposed will help elucidate the impact of spaceflight stress on the production of novel metabolites and is directly responsive to the Space Biology Decadal Survey Recommendation, P1 (Use the ISS as a microbial observatory to determine foundational results that will be critical in the design of strategies to manipulate and control the microbiome of future exploration and contribute to the health of future space explorers). Furthermore, specific project objectives will enable us to answer Space Biology guiding questions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The research will improve our understanding of plant degradation and improvement in the sustainable economy.
Task Progress:	The Decomposition of Complex Organic Matter (DeCOM) project is a ground-based investigation to characterize microorganisms isolated from the ISS in the degradation of plant compost materials. We will utilize various simulated gravity systems available at JPL and NASA to understand the mechanism of how ISS microorganisms assist in the decomposition of complex organic matter during spaceflight. Current efforts will focus on the ground-based analysis for complex sugar/carbohydrate degradation by the fungal and bacterial isolates from the ISS. Objective 1: Identify the ISS microbial species isolated during the MT-1 project which can degrade plant complex organic matter/utilize complex sugar sources: a. Task 1. Selection of the ISS-isolated microbial species capable of growth on a wide range of carbon sources. b. Task 2. Chemical composition of complex sugars associated with plant wastes to resurrect “synthetic” non-degradable sugars. c. Task 3. Analysis of the ISS-isolated microbial species for their ability to degrade 15 different plant waste and generate sugars monomers. Objective 2: Genetically engineer up to five MT-1 microbial species (single or symbiotic combination of bacteria/fungi) down selected from Objective 1 using CRISPR/Cas9 technology for creating mutants that are efficient in degrading complex organic matter. Objective 3: Examine the proteomic profile of filamentous fungi and bacterial mutants created in Objective 2 and wild-type strains using simulated microgravity devices in comparison to ground control. Objective 4: Examine the differential expression of carbohydrate degradation genes and their regulatory genes using simulated microgravity devices in comparison to ground control of mutants and wild-type strains using whole genome sequencing and RNA sequencing. In the second grant period, we made progress on two fronts. Objective 1 was completed. Our analysis identified nine strains capable of growing on all the sugar substrates we tested. Five are downselected for sequencing and downstream experiments. In the next period, complete genome assembly will be completed, and a genetic system is planned to be established for each of the genome sequenced strains. Progress was made on Objective 2. While waiting for the selection of the strains to genetically engineer, we established

	in vitro based CRISPR-Cas9 genetic approaches. We found that the approach is more robust than the in vivo plasmid approach that we developed in the first grant period. We found that the in vitro approach is more robust because we directly protoplast the Cas9 enzyme and synthetic guide RNA. This approach solves the strain variations of the expression of the Cas9 enzymes and guide RNA. Once the genomes sequences are assembled, we will use the revised in vitro approach instead.
Bibliography Type:	Description: (Last Updated: 09/05/2022)
Articles in Peer-reviewed Journals	Bijlani S, Stephens E, Singh NK, Venkateswaran K, Wang CCC. "Advances in space microbiology." iScience. 2021 May 21;24(5):102395. Review. https://doi.org/10.1016/j.isci.2021.102395 ; PMID: 33997680; PMCID: PMC8091056 , May-2021
Articles in Peer-reviewed Journals	Lim S, Bijlani S, Blachowicz A, Chiang YM, Lee MS, Torok T, Venkateswaran K, Wang CCC. "Identification of the pigment and its role in UV resistance in Paecilomyces variotii, a Chernobyl isolate, using genetic manipulation strategies." Fungal Genet Biol. 2021 Jul;152:103567. https://doi.org/10.1016/j.fgb.2021.103567 ; PMID: 33989788 , Jul-2021
Articles in Peer-reviewed Journals	Bijlani S, Singh NK, Eedara VVR, Podile AR, Mason CE, Wang CCC, Venkateswaran K. "Methylobacterium ajmalii sp. nov., isolated from the International Space Station." Front Microbiol. 2021 Mar;12:639396. https://doi.org/10.3389/fmicb.2021.639396 ; PMID: 33790880; PMCID: PMC8005752 , Mar-2021